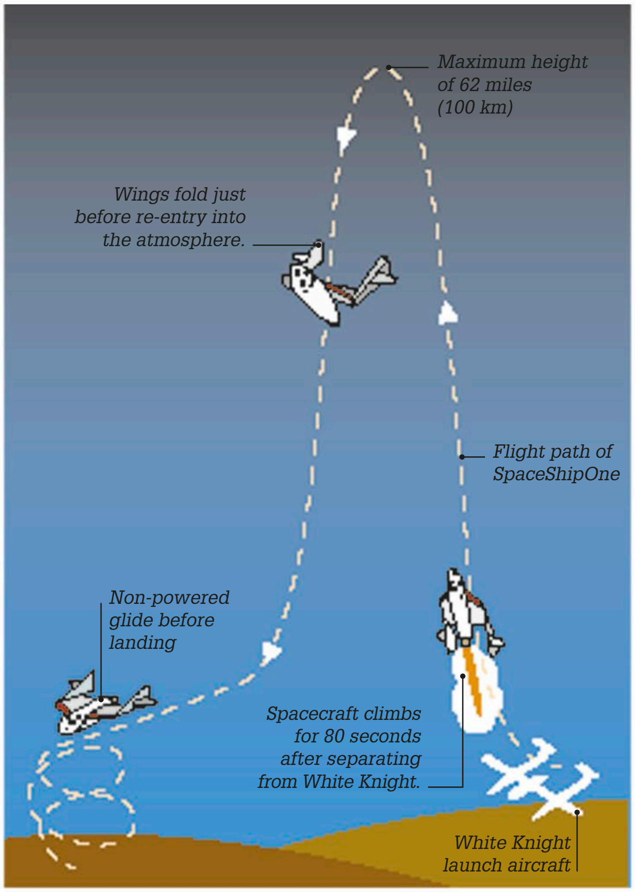
In October 2003, Yang Liwei became China's first astronaut, on the *Shenzhou 5* mission.



SpaceShipOne flight path

2004

First private human spaceflight

SpaceShipOne became the first private spacecraft to travel to the border between the atmosphere and space, at an altitude of 62 miles (100 km) above Earth. The spacecraft could hold up to three crew and passengers, and was carried to 49,212 ft (15,000 m) by the White Knight aircraft (a carrier aircraft designed to launch SpaceShipOne) before firing its own rocket motors for 80 seconds. SpaceShipOne made three flights into space and 17 flights in total.

2004

First brain-computer interface

The first patients were fitted with a prototype of the BrainGate interface—a device that detects brain activity from 96 electrodes implanted into a patient's scalp. The signals are translated by computer into instructions to control a cursor on screen, a robot arm, or a wheelchair.

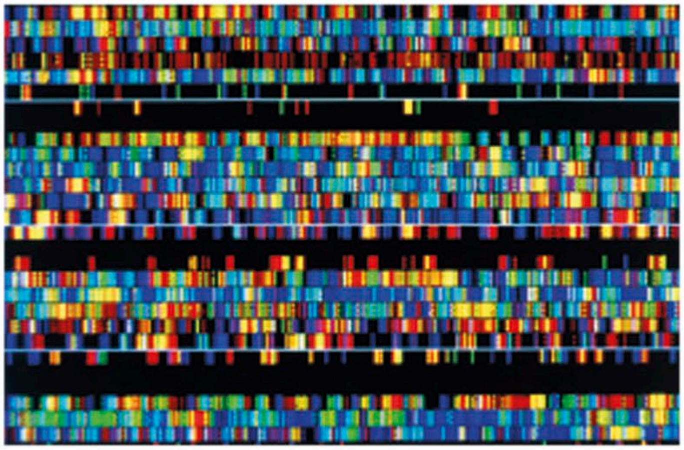
2005

2003

Human Genome Project

The Human Genome Project's completion was announced in 2003. Starting in 1990, this international research effort involved sequencing and mapping the 3.2 billion base pairs (see p.199) that make up all the DNA found in the genes of human beings. Data from the Human Genome Project is used to identify important genes and investigate the genetic causes of certain diseases in order to potentially develop treatments.

The code of life
See pages 198–199

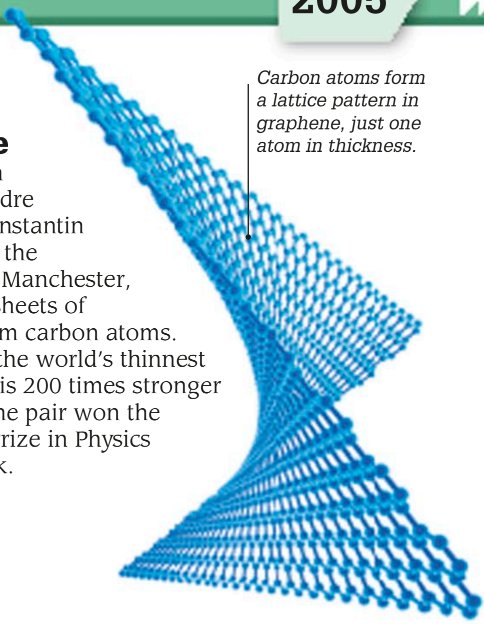


Human DNA sequence displayed as a series of colored bands

2004

Graphene

Russian-born physicists Andre Geim and Konstantin Novoselov at the University of Manchester, UK, created sheets of graphene from carbon atoms. Graphene is the world's thinnest material, yet is 200 times stronger than steel. The pair won the 2010 Nobel Prize in Physics for their work.



> **"With one gram of graphene, you can cover several soccer fields"**
>
> **Andre Geim, on graphene, October 2010**